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Recommendations

International Telecommunication Union
Standardization Sector

Recommendation

ITU-T X.509 (2019) Amd. 1 (10/2024)

SERIES X: Data networks, open system communications and security

Directory

Information technology – Open Systems Interconnection – The Directory: Public-key and attribute certificate frameworks

Amendment 1: Miscellaneous enhancements

ITU-T X-SERIES RECOMMENDATIONS

Data networks, open system communications and security

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INTERNATIONAL STANDARD ISO/IEC 9594-8
RECOMMENDATION ITU-T X.509

Information technology – Open Systems Interconnection – The Directory: Public-key and attribute certificate frameworks

Amendment 1

Miscellaneous enhancements

Summary

Recommendation ITU-T X.509 (2019) | International Standard ISO/IEC 9594-8:2020, Amendment 1 introduces text updates to facilitate separation between cybersecurity ASN.1 modules and directory ASN.1 modules.

History *

Edition	Recommendation	Approval	Study Group	Unique ID
1.0	ITU-T X.509	1988-11-25		11.1002/1000/2999
2.0	ITU-T X.509	1993-11-16	7	11.1002/1000/3000
3.0	ITU-T X.509	1997-08-09	7	11.1002/1000/4123
3.1	ITU-T X.509 (1997) Technical Cor. 1	2000-03-31	7	11.1002/1000/5033
3.2	ITU-T X.509 (1997) Technical Cor. 2	2001-02-02	7	11.1002/1000/5311
3.3	ITU-T X.509 (1997) Technical Cor. 3	2001-10-29	7	11.1002/1000/5559
3.4	ITU-T X.509 (1997) Technical Cor. 4	2002-04-13	17	11.1002/1000/6025
3.5	ITU-T X.509 (1997) Technical Cor. 5	2003-02-13	17	11.1002/1000/6236
3.6	ITU-T X.509 (1997) Technical Cor. 6	2004-04-29	17	11.1002/1000/7285
4.0	ITU-T X.509	2000-03-31	7	11.1002/1000/5034
4.1	ITU-T X.509 (2000) Technical Cor. 1	2001-10-29	7	11.1002/1000/5560
4.2	ITU-T X.509 (2000) Technical Cor. 2	2002-04-13	17	11.1002/1000/6026
4.3	ITU-T X.509 (2000) Technical Cor. 3	2003-02-13	17	11.1002/1000/15258
4.3	ITU-T X.509 (2000) Technical Cor. 3	2004-04-29	17	11.1002/1000/7284
4.4	ITU-T X.509 (2000) Technical Cor. 4	2007-01-13	17	11.1002/1000/8637
5.0	ITU-T X.509	2005-08-29	17	11.1002/1000/8501
5.1	ITU-T X.509 (2005) Cor. 1	2007-01-13	17	11.1002/1000/9051
5.2	ITU-T X.509 (2005) Cor. 2	2008-11-13	17	11.1002/1000/9591
5.3	ITU-T X.509 (2005) Cor. 3	2011-02-13	17	11.1002/1000/11042
5.4	ITU-T X.509 (2005) Cor. 4	2012-04-13	17	11.1002/1000/11577
6.0	ITU-T X.509	2008-11-13	17	11.1002/1000/9590
6.1	ITU-T X.509 (2008) Cor. 1	2011-02-13	17	11.1002/1000/11043
6.2	ITU-T X.509 (2008) Cor. 2	2012-04-13	17	11.1002/1000/11578
6.3	ITU-T X.509 (2008) Cor. 3	2012-10-14	17	11.1002/1000/11736
7.0	ITU-T X.509	2012-10-14	17	11.1002/1000/11735
7.1	ITU-T X.509 (2012) Cor. 1	2015-05-29	17	11.1002/1000/12474
7.2	ITU-T X.509 (2012) Cor. 2	2016-04-29	17	11.1002/1000/12844
7.3	ITU-T X.509 (2012) Cor. 3	2016-10-14	17	11.1002/1000/13032
8.0	ITU-T X.509	2016-10-14	17	11.1002/1000/13031
9.0	ITU-T X.509	2019-10-14	17	11.1002/1000/14033
9.1	ITU-T X.509 (2019) Cor. 1	2021-10-14	17	11.1002/1000/14791
9.2	ITU-T X.509 (2019) Cor. 2	2023-10-29	17	11.1002/1000/15705
9.3	ITU-T X.509 (2019) Amd. 1	2024-10-29	17	11.1002/1000/16172

* To access the Recommendation, type the URL https://handle.itu.int/ in the address field of your web browser, followed by the Recommendation's unique ID.

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INTERNATIONAL STANDARD
ITU-T RECOMMENDATION

Information technology – Open Systems Interconnection – The Directory: Public-key and attribute certificate frameworks

Amendment 1

Miscellaneous enhancements

1) Clause 13.2.13

Add the following clause with a new attribute:

13.2.13 Sequence number attribute type

A value of attribute type **sequenceNumber** is used for holding a sequence number in form of a non-negative integer.

```
sequenceNumber ATTRIBUTE ::= {
  WITH SYNTAX              INTEGER (0..MAX)
  EQUALITY MATCHING RULE   integerMatch
  SINGLE VALUE             TRUE
  LDAP-SYNTAX              integer.&id
  LDAP-NAME                {"Sequence number"}
  ID                       id-at-sequenceNumber }
```

This attribute type may be used to generate an RDN for naming a directory entry holding a public-key or attribute certificate. It is required that the certificate sequence number is non-negative.

2) Annex A, clause A.1

Replace the import section of clause A.1 with:

```
-- A.1 - Authentication framework module

AuthenticationFramework
  {joint-iso-itu-t ds(5) module(1) authenticationFramework(7) 10}
DEFINITIONS ::=
BEGIN

-- EXPORTS All
/*
The types and values defined in this module are exported for use in the other
ASN.1
modules contained within the Directory Specifications, and for the use of other
applications which will use them to access Directory services. Other applications
may
use them for their own purposes, but this will not constrain extensions and
modifications needed to maintain or improve the Directory service.
*/
IMPORTS
  id-asx, id-at, id-ldx, id-lsx, id-mr, id-nf, id-oa, id-oc, id-sc,
  ATTRIBUTE, Attribute{}, commonName, DistinguishedName, Name, NAME-FORM, OBJECT-
CLASS,
  octetString, octetStringMatch, SYNTAX-NAME, top, UnboundedDirectoryString
    FROM UsefulDefinitions
      {joint-iso-itu-t ds(5) module(1) usefulDefinitions(0) 10} WITH SUCCESSORS

/*
  ATTRIBUTE, Attribute{}, DistinguishedName, --distinguishedNameMatch, MATCHING-
RULE, -- Name, NAME-FORM, OBJECT-CLASS,
  -- objectIdentifierMatch, RelativeDistinguishedName, -- SYNTAX-NAME, top
```

```
    FROM InformationFramework
      {joint-iso-itu-t   ds(5)   module(1)   informationFramework(1)   10}   WITH
SUCCESSORS
  --bitStringMatch, boolean, booleanMatch, caseExactMatch, -- commonName,
  -- directoryString, generalizedTime,
  -- generalizedTimeMatch, generalizedTimeOrderingMatch, integer, integerMatch,
  -- integerOrderingMatch, -- octetString, octetStringMatch,
  UnboundedDirectoryString, UniqueIdentifier --, uri
    FROM SelectedAttributeTypes
      {joint-iso-itu-t   ds(5)   module(1)   selectedAttributeTypes(5)   10}   WITH
SUCCESSORS
*/
  algorithmIdentifierMatch, certificateExactMatch, certificateListExactMatch,
  certificatePairExactMatch,      CertificatePoliciesSyntax,      CertPolicyId,
GeneralNames,
  KeyUsage, pkiPathMatch, policyMatch,
  CertificateAssertion, CertificateExactAssertion, CertificateListAssertion,
  CertificateListExactAssertion, CertificatePairAssertion,
  CertificatePairExactAssertion
      FROM CertificateExtensions
        {joint-iso-itu-t  ds(5)  module(1)  certificateExtensions(26)  10}  WITH
SUCCESSORS ;
```

3) Annex A, clause A.2

Replace the import section of clause A.2 with:

```
-- A.2 - Certificate extensions module

CertificateExtensions
  {joint-iso-itu-t ds(5) module(1) certificateExtensions(26) 10}
DEFINITIONS IMPLICIT TAGS ::=
BEGIN

-- EXPORTS ALL

IMPORTS

  id-at, id-ce, id-ldx, id-mr,
  ATTRIBUTE, Attribute{}, MATCHING-RULE, Name, RelativeDistinguishedName, SYNTAX-
NAME,
  UnboundedDirectoryString
    FROM UsefulDefinitions
      {joint-iso-itu-t ds(5) module(1) usefulDefinitions(0) 10} WITH SUCCESSORS
/*
  Name, RelativeDistinguishedName, Attribute{}, MATCHING-RULE,
  SupportedAttributes, SYNTAX-NAME
    FROM InformationFramework
      {joint-iso-itu-t ds(5) module(1) informationFramework(1) 9} WITH SUCCESSORS
*/
  AvlSerialNumber,           CertificateSerialNumber,           CertificateList,
AlgorithmIdentifier{},
  EXTENSION, Time, PolicyID, SupportedAlgorithms
    FROM AuthenticationFramework
      {joint-iso-itu-t   ds(5)   module(1)   authenticationFramework(7)   10}   WITH
SUCCESSORS
/*
  UnboundedDirectoryString
    FROM SelectedAttributeTypes
      {joint-iso-itu-t   ds(5)   module(1)   selectedAttributeTypes(5)   9}   WITH
SUCCESSORS
*/
  ORAddress
    FROM PkiPmiExternalDataTypes
```

```
      {joint-iso-itu-t  ds(5)  module(1)  pkiPmiExternalDataTypes(40)  10}  WITH
SUCCESSORS ;
```

4) Annex A, clause A.3

Replace the import section of clause A.3 with:

```
--  A.3 - Attribute Certificate Framework module

AttributeCertificateDefinitions
  {joint-iso-itu-t ds(5) module(1) attributeCertificateDefinitions(32) 10}
DEFINITIONS IMPLICIT TAGS ::=
BEGIN

-- EXPORTS ALL

IMPORTS

  id-asx, id-at, id-ce, id-mr, id-oc,
  ATTRIBUTE, Attribute{}, AttributeType, AttributeTypeAndValue, directoryString,
  MATCHING-RULE,   OBJECT-CLASS, SYNTAX-NAME, top, TimeSpecification,
  UnboundedDirectoryString
    FROM UsefulDefinitions
      {joint-iso-itu-t ds(5) module(1) usefulDefinitions(0) 10} WITH SUCCESSORS

/*
  The below ASN.1 items have been moved from the InformationFrameword module
  to the UsefulDefinitions module.
  ATTRIBUTE, Attribute{}, AttributeType, AttributeTypeAndValue, MATCHING-RULE,
Name,
  OBJECT-CLASS, RelativeDistinguishedName, SupportedAttributes, SYNTAX-NAME, top
    FROM InformationFramework
      {joint-iso-itu-t ds(5) module(1) informationFramework(1) 9} WITH SUCCESSORS
*/

  ALGORITHM,       AlgorithmIdentifier,       Certificate,       CertificateList,
CertificateSerialNumber,
  EXTENSION, Extensions, InfoSyntax, PolicySyntax, SIGNED{}, SupportedAlgorithms,
  UniqueIdentifier, x509CertificateList
    FROM AuthenticationFramework
      {joint-iso-itu-t  ds(5)  module(1)  authenticationFramework(7)  10}  WITH
SUCCESSORS
/*
  TimeSpecification, UnboundedDirectoryString, UniqueIdentifier
    FROM SelectedAttributeTypes
      {joint-iso-itu-t  ds(5)  module(1)  selectedAttributeTypes(5)  9}  WITH
SUCCESSORS
*/
  certificateListExactMatch, GeneralName, GeneralNames, NameConstraintsSyntax
    FROM CertificateExtensions
      {joint-iso-itu-t  ds(5)  module(1)  certificateExtensions(26)  10}  WITH
SUCCESSORS ;
```

5) Annex B

Replace the import section of Annex B, ***AlgorithmObjectIdentifiers*** *module with:*

```
AlgorithmObjectIdentifiers        {joint-iso-itu-t        ds(5)        module(1)
algorithmObjectIdentifiers(8) 10}
DEFINITIONS ::=
BEGIN
```

```
-- EXPORTS All

/*
The values defined in this module are primarily taking from various specifications
and
collected here for easy reference by other specifcations.

Wen values are copied form an IETF RFC, the IETF RFC number is shown.

When values are copied from the NIST Computer Security Objects Register (CSOR),
the label CSOR is used.
*/

IMPORTS

  algorithm
    FROM UsefulDefinitions
      {joint-iso-itu-t ds(5) module(1) usefulDefinitions(0) 10} WITH SUCCESSORS

  ALGORITHM, AlgorithmIdentifier{}, SupportedAlgorithms, SupportedCurves
    FROM AuthenticationFramework
      {joint-iso-itu-t  ds(5)  module(1)  authenticationFramework(7)  10}  WITH
SUCCESSORS;
```

6) Annex C, clause C.2

Replace the import section of clause C.2 with:

```
ExtensionAttributes {joint-iso-itu-t ds(5) module(1) extensionAttributes(41) 10}
DEFINITIONS ::=
BEGIN

-- EXPORTS All

IMPORTS

  -- from Rec. ITU-T X.501 | ISO/IEC 9594-2

  id-ce, ATTRIBUTE, SYNTAX-NAME
    FROM UsefulDefinitions
      {joint-iso-itu-t ds(5) module(1) usefulDefinitions(0) 10} WITH SUCCESSORS

  -- from Rec. ITU-T X.509 | ISO/IEC 9594-8

  EXTENSION
     FROM AuthenticationFramework
       {joint-iso-itu-t  ds(5)  module(1)  authenticationFramework(7)  10}  WITH
SUCCESSORS

  aAissuingDistributionPoint, altSignatureAlgorithm, altSignatureValue,
  authorityKeyIdentifier,        authorizationValidation,        baseUpdateTime,
basicConstraints,
  certificateIssuer, certificatePolicies, cRLDistributionPoints, cRLNumber,
  cRLStreamIdentifier,    deltaCRLIndicator,    deltaInfo,    expiredCertsOnCRL,
extKeyUsage,
  freshestCRL, holdInstructionCode, invalidityDate, issuerAltName,
  issuingDistributionPoint, keyUsage, nameConstraints, orderedList,
  policyConstraints,    policyMappings,    privateKeyUsagePeriod,    reasonCode,
revokedGroups,
  statusReferrals,           subjectAltName,           subjectAltPublicKeyInfo,
subjectDirectoryAttributes,
  subjectKeyIdentifier, toBeRevoked
     FROM CertificateExtensions
```

```
      {joint-iso-itu-t  ds(5)  module(1)  certificateExtensions(26)  10}  WITH
SUCCESSORS

  acceptableCertPolicies,                          acceptablePrivilegePolicies,
allowedAttributeAssignments,
  attributeDescriptor, attributeMappings, authorityAttributeIdentifier,
  basicAttConstraints, delegatedNameConstraints, groupAC, holderNameConstraints,
  issuedOnBehalfOf, noAssertion, noRevAvail, roleSpecCertIdentifier, singleUse,
  sOAIdentifier, targetingInformation, timeSpecification, userNotice
     FROM AttributeCertificateDefinitions
       {joint-iso-itu-t ds(5) module(1) attributeCertificateDefinitions(32) 10}
WITH SUCCESSORS ;
```

7) Annex D

Replace the import section of Annex D, ***PkiPmiExternalDataTypes module*** *with:*

```
PkiPmiExternalDataTypes         {joint-iso-itu-t         ds(5)         module(1)
pkiPmiExternalDataTypes(40) 10}
DEFINITIONS ::=
BEGIN

-- EXPORTS All

IMPORTS

  intSecurity, PresentationAddress
    FROM UsefulDefinitions {joint-iso-itu-t ds(5) module(1) usefulDefinitions(0)
10}  WITH SUCCESSORS

  EXTENSION
    FROM AuthenticationFramework
      {joint-iso-itu-t  ds(5)  module(1)  authenticationFramework(7)  10}  WITH
SUCCESSORS

  GeneralName
    FROM CertificateExtensions
      {joint-iso-itu-t  ds(5)  module(1)  certificateExtensions(26)  10}  WITH
SUCCESSORS;

/*
  PresentationAddress
    FROM SelectedAttributeTypes
      {joint-iso-itu-t  ds(5)  module(1)  selectedAttributeTypes(5)  10}  WITH
SUCCESSORS
*/
```

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Published in Switzerland
Geneva, 2025